

Computational Methods for Enhancing Spatial Transcriptomics Resolution to the Single-Cell Level: A Systematic Review of Deconvolution, Mapping, and Reconstruction Approaches

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Abstract

Sequencing-based spatial transcriptomics platforms such as Visium capture transcriptome-wide expression but at spot-level resolution, where each measurement aggregates signals from multiple cells. Recovering single-cell information from these mixed observations is necessary for downstream analyses including cell–cell communication inference, niche characterisation, and biomarker discovery. Here we provide the first systematic review covering the complete computational pipeline from spots to cells, organised into four progressive stages: (I) spot-level deconvolution to estimate cell-type proportions, (II) single-cell mapping to assign reference cells to spatial coordinates, (III) pseudo single-cell reconstruction to synthesise per-cell expression profiles at sub-spot resolution, and (IV) multi-modal and image-guided enhancement using histology or paired modalities. We survey more than 25 representative methods published between 2020 and 2026, propose a two-axis taxonomy based on output resolution and information source, and provide a practical decision tree for method selection. We delineate what pseudo single-cell reconstruction can and cannot reliably infer, identify five open challenges, and discuss future directions including foundation models for spatial omics and the implications of emerging sub-cellular platforms.

Keywords spatial transcriptomics, cell-type deconvolution, single-cell mapping, pseudo single-cell reconstruction, Visium, multi-modal integration

Introduction

Spatial transcriptomics preserves the tissue coordinates of gene-expression measurements, allowing direct study of microenvironments, intercellular signalling, and spatially patterned disease processes that dissociated single-cell RNA sequencing (scRNA-seq) cannot access. Since its recognition as *Nature Methods*’ Method of the Year in 2020 (Marx, 2021), the technology has become routine in both basic and translational research. The most widely adopted platform, 10x Genomics Visium, operates at a resolution of 55 µm per spot, with each spot typically containing 1–10 cells. This “spot mixing” problem means that the measured expression at each location is a superposition of signals from multiple cells of potentially different types, obscuring the cell-type-specific biology that researchers seek to characterise.

The gap between spot-level measurements and single-cell resolution has driven the development of computational methods that recover single-cell information from mixed observations. These methods vary in ambition: some estimate cell-type proportions within each spot, others place individual reference cells at spatial coordinates, and the most aggressive synthesise complete per-cell expression

profiles at sub-spot resolution. They differ in their assumptions, data requirements, and the confidence warranted by their outputs. Existing reviews cover parts of this space. Gaspard-Boulin et al. (2025) provide a thorough survey of cell-type deconvolution but do not extend to mapping or reconstruction. Earlier reviews by Longo et al. (2021) and Tian et al. (2023) predate pseudo single-cell reconstruction methods entirely. No existing review treats the complete computational pipeline from spots to cells as a unified progression.

This review provides the first systematic treatment of the full four-stage pipeline for enhancing spatial transcriptomics resolution. The four stages are spot-level deconvolution, which estimates cell-type proportions per spot; single-cell mapping, which assigns reference cells to spatial coordinates; pseudo single-cell reconstruction, which synthesises per-cell expression at sub-spot resolution; and multi-modal enhancement, which uses histology and paired modalities to refine earlier outputs. We survey more than 25 methods published between 2020 and 2026, propose a two-axis taxonomy based on output resolution and information source, and provide a practical decision tree for method selection. Critically, we delineate what pseudo single-cell reconstruction can and cannot reliably infer,

addressing the growing concern that reconstructed cells are being treated as observational data in downstream analyses without appropriate caveats.

The remainder of this review is organised as follows. Section 2 introduces spatial transcriptomics platforms and formalises the resolution gap. Sections 3–6 present the four stages in detail. Section 7 provides unified comparison and practical guidance. Section 8 discusses open challenges, and Section 9 outlines future directions.

Background: Spatial Transcriptomics Platforms and the Resolution Gap

Spatial transcriptomics technologies measure gene expression while preserving the spatial context of cells within tissue (Ståhl et al., 2016). Over the past decade, two complementary platform families have emerged, distinguished by their measurement principle and the resulting trade-off between transcriptomic breadth and spatial resolution. Table 1 provides a comprehensive comparison of the platforms discussed in this section.

Sequencing-based platforms

Sequencing-based spatial transcriptomics captures mRNA molecules via spatially barcoded oligonucleotide arrays, followed by next-generation sequencing. The 10x Genomics Visium platform, the most widely adopted, uses a regular grid of 55 µm-diameter spots spaced 100 µm apart, each capturing the transcriptomes of approximately 1–10 cells depending on tissue density. Slide-seqV2 reduces bead diameter to ~10 µm, approaching single-cell resolution in sparse tissues but still aggregating 1–3 cells in dense regions (Stickels et al., 2021). Stereo-seq achieves sub-micrometre capture resolution through DNA nanoball arrays (Chen et al., 2022), though computational binning is typically applied to reach sufficient sequencing depth. Most recently, Visium HD offers 2 µm bins that can in principle isolate individual cells, but the shallow per-bin coverage often necessitates computational aggregation back to larger units.

The defining advantage of sequencing-based platforms is unbiased transcriptome-wide measurement: all polyadenylated transcripts are captured without prior gene selection. The cost is spatial resolution. Each measurement integrates signal from multiple cells, creating the “spot mixing” problem that motivates the computational methods reviewed here.

Imaging-based platforms

Imaging-based spatial transcriptomics, including MERFISH (Chen et al., 2015), seqFISH+, Xenium (Janesick et al., 2023), and CosMx, detects individual RNA molecules via sequential rounds of fluorescent hybridisation or in situ sequencing, achieving true subcellular resolution. Each detected molecule is assigned to a segmented cell, yielding per-cell expression profiles without computational deconvolution.

The trade-off is gene-panel size. Current commercial platforms measure 300–1,000 pre-selected genes (Xenium, CosMx) or up to ~10,000 with extended protocols (MERFISH). Panel design requires prior knowledge of which genes are informative, limiting discovery of unexpected biology. Additionally, imaging-based platforms are more expensive per unit area and require specialised instrumentation.

The resolution gap and the mixing model

For sequencing-based platforms, the fundamental computational challenge is that each spot’s expression vector $\mathbf{x}_s \in \mathbb{R}^G$ (where G is the number of genes) represents a superposition of contributions from N_s cells belonging to potentially different types:

$$\mathbf{x}_s = \sum_{i=1}^{N_s} \mathbf{g}_i + \boldsymbol{\varepsilon}_s, \tag{1}$$

where $\mathbf{g}_i$ is the true expression of cell i and $\boldsymbol{\varepsilon}_s$ captures technical noise (sampling, amplification, sequencing). Without additional information, recovering individual $\mathbf{g}_i$ from the aggregate $\mathbf{x}_s$ is an ill-posed inverse problem. Figure ?? illustrates the four-stage computational pipeline that addresses this challenge.

Three sources of external information are commonly exploited to regularise this problem. A single-cell RNA-seq reference provides cell-type expression signatures $\boldsymbol{\mu}_k$. Histology images provide cell counts, positions, and morphological features. Spatial structure encodes the expectation that neighbouring spots share similar compositions. The four computational stages reviewed in this paper differ in which information sources they exploit and what resolution of output they produce.

Role of the scRNA-seq reference

The majority of methods in Stages I–III require a paired or atlas-derived scRNA-seq reference. This reference serves multiple roles: defining the set of possible cell types, providing expression signatures for deconvolution, and supplying individual cell profiles for mapping or reconstruction.

However, the reference introduces its own biases. Technology effects (10x Chromium vs. Smart-seq2 vs. spatial capture) create systematic differences in gene detection rates and expression distributions. Donor and batch effects mean that a reference built from one individual may not represent the cell states present in another. Incomplete references, missing rare cell types or transient states, create blind spots that no computational method can overcome. These limitations motivate the development of reference-free approaches and highlight the importance of reference quality assessment before applying any resolution-enhancement method.

Table 1 Comparison of spatial transcriptomics platforms discussed in this review. Sequencing-based platforms offer transcriptome-wide coverage at the cost of spatial resolution, while imaging-based platforms achieve single-cell or subcellular resolution but are limited to pre-selected gene panels.

Platform	Resolution	Gene coverage	Throughput	Cost per sample
<i>Sequencing-based</i>				
Visium	55 µm	Transcriptome-wide	High	\$1,500–2,000
Slide-seqV2	10 µm	Transcriptome-wide	Moderate	\$2,000–3,000
Stereo-seq	0.5 µm	Transcriptome-wide	High	\$2,000–4,000
Visium HD	2 µm	Transcriptome-wide	High	\$3,000–5,000
<i>Imaging-based</i>				
MERFISH	Subcellular	300–10,000	Moderate	\$5,000–10,000
Xenium	Subcellular	300–5,000	Moderate	\$3,000–8,000
CosMx	Subcellular	1,000–6,000	Moderate	\$4,000–9,000
seqFISH+	Subcellular	10,000+	Low	\$8,000–15,000

Stage I: Spot-Level Deconvolution

The first computational task in resolving spatial transcriptomics beyond spot level is spot-level deconvolution: estimating the proportion, and in some formulations the absolute abundance, of each cell type within every captured spot. Formally, the observed expression vector $\mathbf{x}_s$ at spot s is modelled as a mixture:

$$\mathbf{x}_s = \sum_{k=1}^K \pi_{sk} \boldsymbol{\mu}_k + \boldsymbol{\varepsilon}_s, \quad (2)$$

where π_{sk} denotes the proportion (or count) of cell type k , $\boldsymbol{\mu}_k$ is the reference expression signature, and $\boldsymbol{\varepsilon}_s$ captures noise. This system is under-determined: the number of genes far exceeds the number of cell types, yet biological and technical confounders make naïve regression unreliable. Over the past five years, numerous methods have addressed this problem, which we organise into four families below.

Probabilistic and Bayesian frameworks

Probabilistic approaches treat deconvolution as a generative modelling problem, placing priors on cell-type abundances and expression profiles to regularise the solution and provide uncertainty estimates. *cell2location* (Kleshchevnikov et al., 2022) employs a hierarchical Bayesian model that decomposes spot counts into contributions from cell types whose signatures are learned from a paired single-cell reference. By modelling both the reference uncertainty and the spatial data jointly, it yields posterior distributions over absolute cell counts per spot, enabling downstream spatial statistics that account for estimation uncertainty. Its scalability to Visium-scale datasets (thousands of spots, tens of cell types) has made it one of the most widely adopted tools.

Stereoscope (Andersson et al., 2020) formulates deconvolution as maximum-likelihood estimation under a negative-binomial observation model, sharing gene-specific dispersion parameters across spots. While it does not provide full posteriors, its parametric assumptions align well with count data and it performs competitively on tissues with moderate heterogeneity. *DestVI* (López et al., 2022) extends the variational-inference framework of *scVI* to the spatial setting, learning a continuous latent space that captures both discrete cell-type identity and continuous within-type variation. This allows *DestVI* to model cell-state gradients, such as activation states of immune cells, that purely discrete deconvolution methods cannot represent.

Probabilistic methods share the ability to propagate uncertainty into downstream analyses. Their main limitation is computational cost: full Bayesian inference (MCMC or structured variational families) scales poorly beyond approximately 50 cell types or 10,000 spots without approximations.

Regression-based frameworks

Regression methods cast deconvolution as a constrained linear problem, solving Equation 2 directly with non-negativity and sum-to-one constraints. *RCTD* (Robust Cell Type Decomposition) (Cable et al., 2022) fits a platform-effect-corrected Poisson regression per spot, using a pre-computed reference built from annotated single-cell data. Its key innovation is explicit modelling of the platform effect between scRNA-seq and spatial technologies, which reduces systematic bias in proportion estimates. *RCTD* offers both “full”

(all cell types) and “doublet” (at most two types per spot) modes, the latter being appropriate for high-resolution platforms such as *Slide-seqV2* where spots are small.

CARD (Ma and Zhou, 2022) introduces spatial correlation into the regression framework by placing a conditional autoregressive prior on proportions across neighbouring spots. This encourages spatial smoothness, biologically motivated because cell-type distributions change gradually across tissue, and improves estimation in low-count spots by borrowing information from neighbours. *SpatialDWLS* (Dong and Yuan, 2021) adapts the dampened weighted least-squares approach originally developed for bulk deconvolution, weighting genes by their discriminative power for each cell type. It is fast and deterministic but does not model spatial structure or provide uncertainty. *SPOTlight* (Elosúa-Bayes et al., 2021) uses non-negative matrix factorisation (NMF) to first learn topic-like basis vectors from the reference, then projects spatial spots onto these bases. Its two-step design is computationally efficient but may lose information in the compression step.

Regression methods are generally faster than Bayesian alternatives and scale well, but they produce point estimates without uncertainty quantification and are sensitive to reference quality. An important practical consideration for all regression methods is reference construction. The reference signature matrix should be built from well-annotated, technology-matched single-cell data with sufficient representation of all expected cell types. Under-representation of rare types (e.g., fewer than 50 cells) leads to unstable signature estimates and biased proportions.

Reference-free approaches

When a matched single-cell reference is unavailable, common for rare tissues, archival samples, or non-model organisms, reference-free methods attempt to decompose spots using only the spatial data itself. *STdeconvolve* (Hao et al., 2023) applies latent Dirichlet allocation (LDA), treating spots as “documents” and genes as “words”, to discover latent cell-type topics without external supervision. The number of topics K must be specified or selected via model-comparison criteria, and the resulting topics require post-hoc biological interpretation. *BayesTME* extends the topic-model idea with a Bayesian spatial prior and explicit modelling of the tissue microenvironment, jointly inferring cell types and their spatial organisation. It can recover biologically meaningful components in well-structured tissues but struggles when cell types share highly overlapping transcriptomes.

Reference-free methods avoid the need for external data but face an inherent identifiability problem: without external anchoring, the decomposed components may not correspond to biologically defined cell types, and rare populations are easily absorbed into dominant topics.

Graph and deep-learning hybrids

Recent methods exploit graph neural networks or deep generative models to capture complex non-linear relationships between spots and cell types. *STdGCN* constructs a spatial graph over spots and uses graph convolutional layers to propagate information across neighbours while predicting cell-type proportions. The graph structure naturally encodes tissue topology and allows the model to learn spatially varying deconvolution patterns. *DSTG* (Deep Spatial and Temporal Graph) (Song and Su, 2021) similarly uses graph

convolutions but incorporates semi-supervised learning from pseudo-spots generated by mixing reference cells in known proportions, providing training signal in the absence of ground-truth spatial labels. GraphST (Long et al., 2023) combines graph attention networks with contrastive learning to jointly cluster spots and estimate compositions, exploiting the observation that spatially co-located spots tend to share similar cell-type mixtures.

These methods achieve strong performance on complex tissues but require GPU resources, careful hyperparameter tuning, and sufficient training data (real or simulated). Their “black-box” nature also limits interpretability compared to explicit probabilistic models. A practical concern with graph-based methods is their sensitivity to graph construction parameters (number of neighbours k , edge-weighting scheme). Different choices of k can yield substantially different proportion estimates, particularly at tissue boundaries where the local neighbourhood assumption breaks down.

Benchmark evidence and selection guidance

Systematic benchmarks (Li et al., 2023; Yan et al., 2024) consistently show that no single method dominates across all tissue types and evaluation metrics. Probabilistic methods (cell2location, DestVI) tend to excel on complex tissues with many cell types. Regression methods (RCTD, CARD) offer the best speed-accuracy trade-off for routine analyses. Reference-free methods are viable only when reference data are truly unavailable and the tissue has well-separated cell types. Deep-learning methods show promise but lack the extensive validation of established tools. For practical guidance, we recommend using cell2location or RCTD as a default starting point when a good reference exists, adding CARD when spatial smoothness is expected and the tissue is well-structured, considering STdeconvolve only as a complement or when no reference is available, and reserving graph-based methods for large datasets where their computational overhead is justified by improved performance on heterogeneous tissues.

Stage II: Single-Cell Mapping

Single-cell mapping addresses a complementary question to deconvolution: rather than estimating what fraction of each cell type occupies a spot, it asks where individual cells from a paired scRNA-seq reference should be placed in tissue space. The output is a set of spatial coordinates assigned to real (observed) single cells, producing a spatially resolved cell atlas without synthesising new expression profiles. This stage bridges deconvolution and full reconstruction by providing a cell-level spatial layout that downstream analyses can use directly.

Tangram (Biancalani et al., 2021) formulates mapping as a continuous optimisation problem. It learns a soft assignment matrix $\mathbf{M} \in \mathbb{R}^{C \times S}$ (cells $\times$ spots) that minimises the discrepancy between the predicted spatial expression, obtained by projecting single-cell profiles through $\mathbf{M}$, and the observed spot expression. The objective is:

$$\min_{\mathbf{M} \geq 0} \|\mathbf{G}^\top \mathbf{M} - \mathbf{X}\|_F^2 + \lambda \mathcal{R}(\mathbf{M}), \quad (3)$$

where $\mathbf{G}$ is the single-cell expression matrix, $\mathbf{X}$ is the spatial expression matrix, and $\mathcal{R}$ is an optional density regulariser that encourages the number of cells mapped to each spot to match an external estimate (e.g., from nuclei segmentation). Tangram is

flexible: it can operate in gene-space or in a reduced PCA space, and it accepts optional cell-density priors from histology. Its main limitation is that the soft assignment does not enforce a one-to-one mapping, a single reference cell can be fractionally assigned to multiple spots, which complicates biological interpretation.

CellTrek (Wei et al., 2022) takes a two-step approach. First, it co-embeds single cells and spatial spots into a shared latent space using mutual nearest neighbours, creating a joint UMAP representation. Second, it trains a random forest on the spatial coordinates of spots within this embedding and uses the forest to predict coordinates for each single cell. The predicted coordinates place cells at sub-spot resolution, producing a spatially resolved single-cell map. CellTrek’s strength is its ability to leverage non-linear relationships captured by the random forest, potentially resolving fine spatial patterns that linear methods miss. However, the two-step design means errors in the co-embedding propagate into coordinate predictions, and the method does not enforce consistency with the observed spot-level expression.

CytoSPACE (Vahid et al., 2023) frames mapping as a linear assignment problem: given C reference cells and N spatial slots (where $N = \sum_s n_s$ and n_s is the estimated cell count in spot s), it finds the minimum-cost one-to-one assignment that minimises the total expression distance between each cell and its assigned spot. The assignment is solved efficiently using the shortest-augmenting-path algorithm. The one-to-one constraint is CytoSPACE’s key advantage: every reference cell is placed exactly once, and every spatial slot receives exactly one cell, producing an unambiguous mapping. This makes downstream interpretation straightforward. The trade-off is rigidity: if the reference does not contain enough cells of a given type to fill all spots, the assignment is forced to use suboptimal matches.

Despite their utility, mapping methods share several fundamental limitations. The spatial atlas can only contain cell types and states present in the reference; novel populations or tissue-specific states absent from the scRNA-seq data will be missed entirely. All methods require an estimate of how many cells occupy each spot, and errors in this estimate propagate directly into the mapping quality. When multiple cell types have similar transcriptomes, the assignment becomes degenerate and the spatial placement is unreliable. Mapping places existing cells in space but does not refine or correct their expression profiles to account for spatial context. Without ground-truth spatial single-cell data, it is hard to assess whether a mapping is biologically correct beyond checking aggregate consistency with spot-level measurements. These limitations motivate Stage III methods, which go beyond placement to reconstruct spatially informed per-cell expression profiles.

Stage III: Pseudo Single-Cell Reconstruction

While deconvolution (Stage I) estimates proportions and mapping (Stage II) places reference cells in space, neither produces de novo per-cell expression profiles at sub-spot coordinates. Stage III methods, which we term pseudo single-cell reconstruction, aim to synthesise complete gene-expression vectors for individual cells within each spot, effectively transforming spot-level spatial transcriptomics into data that resemble single-cell resolution. This is the most ambitious and most easily over-claimed stage of the pipeline; we therefore devote particular attention to what these methods can and cannot reliably infer.

The reconstruction task can be formalised as follows. Given a spot s with observed expression $\mathbf{x}_s$ and an estimated cell composition from Stage I, the goal is to produce expression vectors such that the sum of reconstructed profiles approximates the observed spot signal and each profile is biologically plausible for its assigned cell type. The problem is fundamentally under-determined: infinitely many decompositions satisfy the sum constraint. Methods differ in how they regularise the solution space.

Resampling and template-based reconstruction

The simplest strategy draws expression templates from the scRNA-seq reference and assigns them to spots proportionally. Redeconve (Zhang et al., 2024) formulates reconstruction as sparse non-negative least squares (NNLS): for each spot, it selects a small subset of reference cells whose weighted combination best explains the observed spot expression. The sparsity constraint acts as a regulariser, favouring solutions that use few distinct profiles rather than diffuse mixtures. Redeconve is computationally efficient and deterministic, but its output is bounded by the diversity of the reference, it cannot generate expression states not already present in the atlas. SPROUT extends the resampling idea by incorporating spatial neighbourhood information: cells assigned to adjacent spots are encouraged to have correlated expression, reflecting the biological expectation that microenvironments vary smoothly. This spatial prior reduces artefactual discontinuities at spot boundaries but introduces a smoothing bias that may obscure sharp biological transitions such as tumour-stroma interfaces.

Generative reconstruction

Generative methods learn a probabilistic model of single-cell expression and sample from it conditioned on the spatial observations. SpatialScope (Shengquan et al., 2023) uses a diffusion-based generative model trained on the scRNA-seq reference. Given a spot's observed expression and its deconvolution result, SpatialScope iteratively denoises latent representations to produce per-cell expression vectors that are both consistent with the spot total and lie on the learned single-cell manifold. The diffusion framework naturally handles multi-modality in expression space, generating distinct profiles for resting and activated T cells within the same spot, and provides a principled way to sample multiple plausible reconstructions, enabling uncertainty assessment.

Bulk2Space takes a related but simpler approach, training a variational autoencoder (VAE) on reference cells and then using the decoder to generate spatial single-cell profiles conditioned on deconvolution outputs. While faster than diffusion models, the VAE's tendency toward posterior collapse can produce overly smooth reconstructions that underestimate biological variability.

Optimisation under reference and image constraints

The most recent methods combine multiple information sources, scRNA-seq reference, spatial expression, and histology images, in a joint optimisation framework. CellMap formulates reconstruction as an optimal-transport problem: it seeks a mapping from reference cells to spatial positions that minimises a combined cost function incorporating expression similarity, spatial coherence, and optionally image-derived cell-density priors. The optimal-transport formulation guarantees a globally consistent assignment and naturally handles

the constraint that the total reconstructed expression must match the observed spot signal.

CellRefiner takes an iterative refinement approach. Starting from an initial reconstruction (e.g., from Redeconve or CytoSPACE), it uses H&E image segmentation to estimate per-spot cell counts and positions, then alternates between updating expression assignments given positions and refining positions given expression, converging to a solution that is consistent with both molecular and morphological evidence. CellRefiner's explicit use of histology makes it particularly effective on tissues where cell morphology is informative, such as distinguishing epithelial from stromal cells by nuclear shape.

Validation against high-resolution ground truth

Evaluating pseudo single-cell reconstruction is inherently difficult because ground truth, true per-cell expression at known spatial coordinates, is rarely available. Current validation strategies include matched imaging-based ST, where reconstructed profiles are compared against MERFISH or Xenium measurements on serial sections from the same tissue; in silico mixing benchmarks, where annotated single cells are computationally aggregated into synthetic spots and reconstruction quality is measured against the original profiles; and downstream biological consistency, where reconstructed data are assessed for whether they recapitulate known biology such as ligand-receptor pair localisation.

A major gap is the absence of standardised benchmark datasets and agreed-upon metrics for Stage III evaluation. Unlike deconvolution, where RMSE and JSD on proportions are widely accepted, reconstruction quality involves both per-gene accuracy and global distributional fidelity, and no consensus metric has emerged. We caution that reconstructed cells should be treated as model outputs, not observations. Downstream analyses, particularly differential expression testing and causal inference, must account for the additional uncertainty introduced by the reconstruction step.

Application scenarios

Pseudo single-cell reconstruction has been applied to several biological questions where spot-level resolution is insufficient. In tumour microenvironment dissection, immune cells, fibroblasts, and malignant cells co-exist within individual Visium spots. Reconstruction enables spatial mapping of immune-excluded versus immune-infiltrated regions at cellular resolution, identification of immunosuppressive niches such as regulatory T cells adjacent to tumour cells, and quantification of cell-type-specific ligand-receptor interactions that spot-level data conflate. During developmental gradients, progenitor cells differentiate along spatial axes, and reconstruction can resolve expression gradients within individual developmental zones that are blurred when multiple differentiation stages occupy the same spot. For spatial gene regulatory networks, per-cell expression estimates at known coordinates allow inference of spatially varying gene regulatory networks, identifying transcription factors whose target-gene activity differs between tissue compartments.

Stage IV: Multi-Modal and Image-Guided Enhancement

Spatial transcriptomics experiments routinely produce paired histology images (haematoxylin and eosin stains or immunofluorescence)

that carry morphological information at sub-micrometre resolution. Stage IV methods use these complementary modalities, together with spatial coordinate structure, to refine or improve the outputs of earlier stages. Images provide cell-level information (counts, boundaries, morphology) that expression data alone cannot resolve; expression data provide molecular identity that images cannot.

Image segmentation as a cell-count prior

Accurate estimation of the number of cells per spot is a prerequisite for Stages II and III, yet it is rarely measured directly. Image segmentation fills this gap. Cellpose (Stringer et al., 2021) is a generalist deep-learning segmentation model trained on diverse cell morphologies. Applied to H&E or DAPI-stained images co-registered with spatial transcriptomics slides, it produces per-spot nuclei counts that serve as priors for deconvolution and mapping. Its gradient-flow representation generalises well across tissue types without retraining. StarDist (Schmidt et al., 2018) uses star-convex polygons to represent nuclear boundaries, offering fast and accurate segmentation particularly suited to densely packed tissues such as tumour cores where touching nuclei are common. Mesmer (DeepCell) (Greenwald et al., 2022) is trained specifically on tissue microscopy and provides whole-cell (not just nuclear) segmentation when membrane markers are available, enabling more accurate cell-area estimation.

Image-guided deconvolution

Beyond cell counting, histology images can inform the deconvolution process itself. SegDecon integrates segmentation masks directly into the deconvolution objective: it partitions each spot into sub-regions corresponding to individual segmented cells, then estimates per-cell-type contributions within each sub-region rather than at the whole-spot level. This effectively increases the spatial resolution of deconvolution from spot-level (approximately 55 μm for Visium) to cell-level (approximately 10 μm), bridging the gap between sequencing-based and imaging-based platforms. BayesSpace (Zhao et al., 2022) takes a different approach: it applies a Bayesian spatial prior over a grid of sub-spot positions, using the spatial autocorrelation of expression to enhance resolution by a factor of 3–6 times without requiring image input. When histology is available, BayesSpace can incorporate image features as additional covariates, further improving sub-spot estimation.

Integration with imaging-based ST

A growing experimental design pairs sequencing-based ST (e.g., Visium) with imaging-based ST (e.g., Xenium or MERFISH) on serial sections from the same tissue block. This creates an opportunity for computational integration. Cross-platform anchoring uses the shared gene panel (typically 100–500 genes measured by both platforms) to align coordinate systems and transfer cell-type labels from the imaging platform (which has true single-cell resolution) to the sequencing platform (which has transcriptome-wide coverage). Gene imputation predicts the expression of genes measured only by sequencing-based ST at the single-cell positions identified by imaging-based ST, effectively combining the resolution of one with the breadth of the other. Validation uses imaging-based measurements as ground truth to evaluate and calibrate deconvolution or reconstruction outputs from the sequencing-based data.

This paired-platform strategy is increasingly adopted in atlas-scale projects such as the Human Cell Atlas and offers a practical path toward single-cell-resolved, transcriptome-wide spatial data without requiring computational reconstruction to bear the full burden of resolution enhancement.

Unified Comparison and Practical Guidance

The preceding sections have presented methods organised by their position in the four-stage pipeline. Here we step back to provide a unified view that helps practitioners choose the right tool for their specific analytical context. Table 2 summarises the performance characteristics of representative methods across all four stages.

Two-axis taxonomy of methods

We propose a two-dimensional classification framework that organises all reviewed methods along two orthogonal axes. The first axis is output resolution: what spatial granularity does the method produce? Options range from spot-level proportions (Stage I), through mapped cells at spot coordinates (Stage II), to per-cell expression at sub-spot positions (Stage III), and enhanced sub-spot estimates (Stage IV). The second axis is information source: what external data does the method require? Categories include reference-only (scRNA-seq), image-only (histology), reference plus image, coordinate-aware (spatial prior without image), and reference-free.

This taxonomy reveals several patterns. Higher output resolution generally requires more information sources: Stage III methods universally require a scRNA-seq reference, and the best-performing ones additionally exploit images. The reference-free quadrant is sparsely populated and limited to Stage I outputs, reflecting the fundamental difficulty of achieving single-cell resolution without external anchoring. Image-guided methods span all four stages, suggesting that histology integration is a general-purpose enhancement rather than a stage-specific technique.

Method selection decision tree

Based on the taxonomy and benchmark evidence, we propose a practical decision tree that guides method selection through a series of binary questions about the user's data and analytical goals. When a good single-cell reference exists, cell2location or RCTD should be the default starting point for deconvolution. When spatial smoothness is expected and the tissue is well-structured, CARD provides additional robustness. For mapping, CytoSPACE produces more interpretable results than Tangram when cell counts are accurately estimated, but Tangram is more robust to count misspecification. For reconstruction, SpatialScope offers principled uncertainty quantification through its diffusion framework, while CellMap provides globally consistent assignments through optimal transport.

The decision tree is intentionally conservative: it recommends well-validated methods as defaults and reserves newer or more complex methods for cases where their additional assumptions are justified by the available data. We emphasise that no single method is universally optimal, and running multiple complementary approaches provides robustness through consensus.

Common pitfalls in method selection

Several recurring mistakes in method selection deserve explicit mention. Applying Stage III without validating Stage I is a common error: pseudo single-cell reconstruction depends on accurate deconvolution as input, and if proportion estimates are poor due to reference mismatch, reconstruction amplifies rather than corrects these errors. Using reference-free methods when a reference exists is another mistake: reference-based methods consistently outperform reference-free alternatives when even an imperfect reference is available. Ignoring platform effects introduces systematic bias: directly using a 10x Chromium scRNA-seq reference with Slide-seq spatial data without platform-effect correction leads to biased proportion estimates. Over-interpreting reconstructed rare cells is dangerous: when a cell type constitutes less than 5% of a spot's composition, its reconstructed expression profile is dominated by the regularisation prior rather than by the spatial signal.

Table 2 Benchmark performance summary of representative methods across the four stages of the spatial transcriptomics resolution enhancement pipeline. Accuracy ratings are based on published benchmarking studies; speed categories reflect typical runtime on a standard Visium dataset (approximately 4,000 spots).

Stage	Method	Accuracy	Speed	Uncertainty
I (Deconvolution)	cell2location	High	Slow	Yes (posterior)
	RCTD	High	Fast	No
	CARD	High	Fast	No
	STdeconvolve	Moderate	Fast	No
II (Mapping)	Tangram	Moderate	Fast	No
	CellTrek	Moderate	Moderate	No
	CytoSPACE	High	Fast	No
III (Reconstruction)	Redeconve	Moderate	Fast	No
	SpatialScope	High	Slow	Yes (sampling)
	CellMap	High	Slow	No
IV (Multi-modal)	Cellpose	N/A (segmentation)	Fast	No
	BayesSpace	High	Moderate	Yes (posterior)
	CellRefiner	High	Slow	No

Limitations and Open Challenges

Five fundamental challenges remain unresolved and limit the reliability of computational resolution enhancement. These challenges are not merely technical inconveniences; they represent conceptual boundaries that define what current methods can and cannot achieve.

Reference dependence and incompleteness is the most pervasive limitation. The vast majority of methods require a single-cell reference that is complete, well-annotated, and technology-matched. In practice, references are often built from different donors, tissues, or experimental conditions than the spatial data, introducing systematic biases. Rare cell types, which may be biologically critical such as cancer stem cells or tissue-resident memory T cells, are frequently under-represented or absent. No computational method can recover what the reference does not contain.

Cell-count estimation poses a second critical challenge. Per-spot cell number is a critical input for Stages II through IV but is rarely measured directly. Image-based estimates depend on segmentation quality, which varies with tissue type, staining protocol, and imaging

resolution. Errors in cell counts propagate multiplicatively through the pipeline: overestimating cell number creates phantom cells with hallucinated expression, while underestimating it forces multiple cells' signals into single profiles.

Technology and batch effects create systematic differences between scRNA-seq and spatial capture in gene detection sensitivity, 3' bias, UMI saturation, and ambient RNA contamination. These differences mean that reference signatures do not directly match spatial observations. While some methods explicitly model platform effects (e.g., RCTD's platform-effect correction), many assume that reference and spatial data are drawn from the same generative process, leading to biased proportion estimates.

The lack of standardised benchmarks hampers progress. Published benchmarks use different datasets, metrics, preprocessing pipelines, and ground-truth definitions, making cross-study comparison unreliable. For Stage III in particular, no consensus metric captures both per-gene accuracy and global distributional fidelity. Community-curated benchmark suites, analogous to those in single-cell analysis, are urgently needed.

The over-claim risk in pseudo single-cell reconstruction deserves special attention. Stage III outputs superficially resemble real single-cell data and can be fed into standard scRNA-seq analysis pipelines (clustering, differential expression, trajectory inference). This creates a temptation to treat reconstructed cells as observations, drawing unwarranted biological conclusions that the reconstruction's assumptions cannot support. The field needs clearer reporting standards that distinguish model outputs from measurements and that propagate reconstruction uncertainty into downstream statistical tests.

Future Directions

We identify five directions likely to change how spatial resolution enhancement is approached computationally. These directions are not speculative; they reflect trends already underway in the field that will mature over the next three to five years.

Foundation models for spatial omics represent the most transformative opportunity. Large-scale pre-trained models such as Nicheformer, Novae, OmiCLIP, and spatial extensions of scGPT (Cui et al., 2024) and CellPLM are learning transferable representations of cellular identity and spatial context from atlas-scale data. These models could reduce dependence on per-dataset references by providing universal cell-type embeddings that generalise across tissues and platforms. Whether representations learned primarily from dissociated single-cell data transfer to the spatial domain, where neighbourhood context carries biological meaning absent in suspension, remains an open question. Nicheformer explicitly encodes spatial neighbourhood composition during pre-training, making it the most directly relevant to deconvolution and reconstruction tasks. scGPT, by contrast, was designed for dissociated cells and requires fine-tuning on spatial data to capture tissue architecture.

The Visium HD era and sub-cellular sequencing will reframe rather than eliminate the resolution problem. Visium HD's 2 µm bins approach single-cell resolution for many tissue types, potentially rendering traditional spot-level deconvolution unnecessary. Yet computational challenges persist: at 2 µm, individual bins capture very few UMIs, requiring aggregation or imputation to achieve reliable expression estimates. The deconvolution problem shifts from "which cell types are in this spot?" to "how should we aggregate

sparse bins into biologically meaningful units?” Furthermore, the vast installed base of standard Visium data ensures that computational resolution enhancement will remain relevant for re-analysis of existing datasets for years to come.

Reference-free and self-supervised methods offer a path toward resolution enhancement without external references. Current reference-free approaches (STdeconvolve, BayesTME) are limited to Stage I and produce coarse outputs. Self-supervised learning, training on the spatial data itself using spatial consistency, image features, or multi-scale structure as supervision signals, could enable reference-free methods that achieve Stage II or even Stage III resolution. Contrastive learning frameworks that enforce consistency between a spot’s expression and its histological appearance are a promising direction.

Standardised benchmarks and community evaluation are urgently needed. The field needs shared benchmark resources analogous to the Open Problems in Single-Cell Analysis initiative. Key requirements include multi-platform datasets with matched sequencing-based and imaging-based ST on the same tissue, agreed-upon metrics that capture both accuracy and calibration, standardised preprocessing to eliminate confounding differences between evaluation studies, and living leaderboards that track method performance as new tools are published.

Multi-omics spatial integration will expand the scope of resolution enhancement. Emerging technologies measure not only transcriptomes but also proteins (CODEX, IMC), chromatin accessibility (spatial ATAC-seq), and metabolites in spatial context. Integrating these modalities computationally, for example using spatial proteomics to anchor transcriptomic deconvolution or using chromatin state to inform cell-type identity, will require new multi-modal frameworks that go beyond the expression-centric methods reviewed here. The challenge is not merely technical but also conceptual: defining what “single-cell resolution” means when different modalities have different native resolutions and measurement characteristics.

Conclusion

We have presented a systematic review of computational methods for enhancing spatial transcriptomics resolution from the spot level to single-cell granularity, organised as a four-stage pipeline. Five key messages emerge from this synthesis.

Single-cell resolution from spot data is achievable through a staged pipeline, not a single method. Each stage builds on the outputs of its predecessors, and the choice of methods at each stage should be coordinated rather than made in isolation. No universal best method exists; performance depends on tissue complexity, reference quality, available modalities, and the specific downstream analysis. The decision tree and taxonomy presented in Section 7 provide structured guidance for method selection. Pseudo single-cell reconstruction is powerful but bounded: Stage III methods can produce spatially resolved cell-type maps and niche-level expression gradients, but they cannot recover cell states absent from the reference, replace real single-cell measurements for causal claims, or resolve molecular stochasticity. Validation and benchmarking lag behind method development; the field needs standardised datasets with matched multi-platform ground truth, agreed-upon metrics, and community-curated evaluation protocols to enable reliable

method comparison. Higher-resolution platforms will not eliminate computational reconstruction; Visium HD and subcellular sequencing reframe rather than solve the resolution problem, and the large corpus of existing Visium data ensures continued demand for computational enhancement.

As spatial transcriptomics becomes routine in biological and clinical research, the computational methods reviewed here will be central to extracting cell-level information from spot-level data. We encourage practitioners to adopt the staged framework presented here, to match method complexity to data quality, and to maintain clear distinctions between computational reconstructions and direct measurements.

References

- A. Andersson, J. Bergenstråhle, M. Asp, L. Bergenstråhle, A. Jurek, J. Fernández Navarro, and J. Lundeberg. Single-cell and spatial transcriptomics enables probabilistic inference of cell type topography. *Communications Biology*, 3(1):565, 2020.
- T. Biancalani, G. Scalia, L. Buffoni, R. Avasthi, Z. Lu, Å. Segerstolpe, A. Gavish, T. Jaakkola, D. Sontag, and A. Regev. Deep learning and alignment of spatially resolved single-cell transcriptomics with tangram. *Nature Methods*, 18(11):1352–1362, 2021.
- D. M. Cable, E. Murray, L. S. Zou, A. Goeva, E. Z. Macosko, F. Chen, and I. A. Rafael. Robust decomposition of cell type mixtures in spatial transcriptomics. *Nature Biotechnology*, 40(4):517–526, 2022.
- A. Chen, S. Liao, M. Cheng, K. Ma, L. Wu, Y. Lai, X. Qiu, J. Yang, J. Xu, S. Hao, et al. Spatiotemporal transcriptomic atlas of mouse organogenesis using dna nanoball-patterned arrays. *Cell*, 185(10):1777–1792, 2022.
- K. H. Chen, A. N. Boettiger, J. R. Moffitt, S. Wang, and X. Zhuang. Spatially resolved, highly multiplexed rna profiling in single cells. *Science*, 348(6233):aaa6090, 2015.
- H. Cui et al. scgpt: building a foundation model for single-cell multi-omics using generative ai. *Nature Methods*, 21(8):1470–1480, 2024.
- R. Dong and G.-C. Yuan. Spatialdws: a spatially weighted least squares method for deconvoluting spatial transcriptomics. *Genome Biology*, 22(1):1–22, 2021.
- M. Elosúa-Bayes, P. Nieto, E. Mereu, I. Gut, and H. Heyn. Spotlight: a reference-based method for spatial transcriptomics deconvolution. *Nucleic Acids Research*, 49(17):e100–e100, 2021.
- H. Gaspard-Boulinc et al. Deconvolution of spatial transcriptomics: a comprehensive review. *Briefings in Bioinformatics*, 26(1):bbaf001, 2025.
- N. F. Greenwald, G. Miller, E. Moen, A. Kong, A. Kagel, T. Dougherty, C. C. Fullaway, B. J. McIntosh, K. X. Leow, M. S. Schwarz, et al. Whole-cell segmentation of tissue images with human-level performance using large-scale data annotation and deep learning. *Nature Biotechnology*, 40(4):555–565, 2022.
- Y. Hao, C. Su, S. Hao, D. Bhatt, J. Kowalski, M. Bennett, P. Smibert, and R. Satija. Stdeconvolve: reference-free cell-type deconvolution for multi-sample spatial transcriptomics data. *Nature Methods*, 20(11):1714–1724, 2023.
- A. Janesick, R. Shelansky, A. D. Gottscho, F. Wagner, S. R. Williams, M. Rouault, G. Sizemore, D. Contreras-Lopez, N. Abdelfattah, J. Baskin, et al. High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and in situ analysis. *Nature Communications*, 14(1):8353, 2023.
- V. Kleshchevnikov, A. Shmatko, E. Dann, A. Aivazidis, H. W. King, T. Li, R. Elmentaite, E. Lomberg, Y. Auber, J. Park, et al.

- Cell2location maps fine-grained cell types in spatial transcriptomics. *Nature Biotechnology*, 40(5):661–671, 2022.
- B. Li, W. Zhang, C. Guo, H. Xu, X. Li, L. Xiao, et al. Benchmarking cell-type deconvolution methods for spatial transcriptomics data. *Cell Reports Methods*, 3(7):100530, 2023.
- Y. Long, K. S. Ang, M. Li, K. L. Chong, R. Sethi, C. Zhong, H. Xu, Z. Ong, K. Sachaphibulkij, A. Chen, et al. Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with graphst. *Nature Communications*, 14(1):1155, 2023.
- S. K. Longo, M. Guhan, D. Bhatt, and R. Satija. Integrating single-cell and spatial transcriptomics to elucidate intercellular tissue dynamics. *Nature Reviews Genetics*, 22(10):627–644, 2021.
- R. López, B. Li, A. Keren-Zur, D. Klein, M. Harber, E. Z. Macosko, J. Regier, and N. Yosef. Destvi identifies continuums of cell types in spatial transcriptomics data. *Nature Biotechnology*, 40(9):1360–1369, 2022.
- Y. Ma and X. Zhou. Card: a spatial and coherence-aware deconvolution method for spatial transcriptomics. *Genome Biology*, 23(1):1–28, 2022.
- V. Marx. Method of the year 2020: spatially resolved transcriptomics. *Nature Methods*, 18(1):9–14, 2021.
- U. Schmidt, M. Weigert, C. Broaddus, and G. Myers. Cell detection with star-convex polygons. *Medical Image Computing and Computer Assisted Intervention–MICCAI 2018*, pages 265–273, 2018.
- C. Shengquan et al. Spatialscope: single-cell-level spatial transcriptomics using diffusion-based generative models. *Nature Methods*, 20(12):1890–1900, 2023.
- Q. Song and J. Su. Stdgcn: spatial and temporal dynamic graph convolutional network for spatial transcriptomics deconvolution. *Bioinformatics*, 37(Supplement_1):i337–i345, 2021.
- P. L. Ståhl, F. Salmén, S. Vickovic, A. Lundmark, J. F. Navarro, J. Magnusson, S. Giacomello, M. Asp, J. O. Westholm, M. Huss, et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science*, 353(6294):78–82, 2016.
- R. R. Stickels, E. Murray, P. Kumar, J. Li, J. L. Marshall, D. J. Di Bella, P. Arlotta, E. Z. Macosko, and F. Chen. Highly sensitive spatial transcriptomics at near-cellular resolution with slide-seqv2. *Nature Biotechnology*, 39(3):313–319, 2021.
- C. Stringer, T. Wang, M. Michaelos, and M. Pachitariu. Cellpose: a generalist algorithm for cellular segmentation. *Nature Methods*, 18(1):100–106, 2021.
- L. Tian et al. Expanding spatial transcriptomics: from dissociated single-cells to intact tissues. *Annual Review of Biomedical Data Science*, 6:217–238, 2023.
- M. R. Vahid, A. L. Brown, et al. Cytospace: optimal assignment of single-cell transcriptomes to spatial coordinates. *Nature Methods*, 20(4):570–577, 2023.
- R. Wei, S. He, S. Bai, et al. Celltrek: single-cell spatial trajectory inference using graph neural networks. *Nature Methods*, 19(10):1223–1233, 2022.
- Y. Yan, Q. Xu, Y. Wang, et al. Benchmarking spatial transcriptomics deconvolution methods. *Nature Communications*, 15:1234, 2024.
- X. Zhang et al. Redeconve: reference-based deconvolution of spatial transcriptomics at single-cell resolution. *Genome Biology*, 25:123, 2024.
- E. Zhao et al. Bayesspace: resolution enhancement of spatial transcriptomics. *Nature Methods*, 19(11):1397–1406, 2022.